

Analog Audio Amplifier Design: Differential Pre-Amplifier, CE Gain Stage, Active Band-Pass Filter, and Class-AB Power Amplifier

Nav Singhal
Roll No.: 2024102004
IIIT Hyderabad
Hyderabad, India

Email: (nav.singhal@students.iiit.ac.in)

Ravi Kumar Sahu
Roll No.: 2024102024
IIIT Hyderabad
Hyderabad, India

Email: (ravi.sahu@students.iiit.ac.in)

Abstract—This report presents the design and analysis of an analog audio amplifier meeting audible-band specifications using a cascaded architecture: a differential pre-amplifier for noise rejection and initial gain, a common-emitter (CE) gain stage for high voltage amplification, an active band-pass filter for 20 Hz–20 kHz passband shaping, and a class-AB power amplifier for driving a low-resistance load with approximately unity voltage gain. Theoretical derivations are provided using symbolic variables, alongside analysis metrics such as gain, input/output impedances, common-mode rejection ratio (CMRR), power delivery, total harmonic distortion (THD), and slew-rate constraints.

Index Terms—audio amplifier, differential amplifier, common-emitter, active filter, band-pass, class-AB, CMRR, THD, slew rate

I. INTRODUCTION

The goal is to design an audio amplifier that satisfies typical project requirements (gain, bandwidth, load drive, and stage-wise constraints) and evaluation criteria for differential pre-amplifier, CE gain stage, filter stage, power amplifier stage, and overall performance analysis.

II. SYSTEM OVERVIEW AND SPECIFICATIONS

A. Stage-wise architecture

We implement a four-stage chain:

- 1) Differential pre-amplifier (initial gain + common-mode noise rejection)
- 2) CE gain stage (major voltage gain contribution)
- 3) Active band-pass filter (passband: f_L to f_H)
- 4) Class-AB power amplifier (high current drive, $\approx$ unity voltage gain)

B. Symbols and definitions

Let $v_{in}(t)$ be the small-signal input, $v_o(t)$ the final output, and the total midband voltage gain be

$$A_v^{\text{total}} = A_{v1} A_{v2} A_{v3} A_{v4}. \quad (1)$$

Thermal voltage is V_T , BJT current gain is β , transconductance is $g_m = I_C/V_T$, and $r_\pi \approx \beta/g_m$.

(Optional) Identify funding agency here. If none, delete this.

C. Design Constraints

The amplifier was designed subject to the following practical constraints typically encountered in audio amplifier design and in the laboratory environment.

- **Supply rails:** The amplifier operates from symmetric supply rails of $\pm 5\text{ V}$. This limits the maximum achievable output swing and therefore constrains the maximum deliverable power.
- **Target frequency range:** The system was designed to operate across the audible audio band of approximately 20 Hz to 20 kHz.
- **Voltage gain requirement:** The amplifier must provide sufficient overall voltage gain to amplify small input signals to a level capable of driving a low-resistance load.
- **Low-resistance load drive:** The final stage must be capable of delivering current to a load of approximately $10\ \Omega$.
- **Stage-wise architecture requirement:** The design must include the following stages as specified in the project requirements:
 - 1) Differential pre-amplifier
 - 2) Common-emitter voltage gain stage
 - 3) Active band-pass filter
 - 4) Class-AB power amplifier
- **Low distortion requirement:** The amplifier should maintain reasonably low total harmonic distortion across the operating range.
- **Practical component constraints:** The design must use standard laboratory components and commercially available BJTs, resistors, capacitors, and diodes.

III. FULL CIRCUIT SCHEMATIC AND PHYSICAL CIRCUIT

Referring to Fig. 1, on the top right we have the pre-amp stage, moving to the top left we have the CE amp stage. In the middle left we have the 7905 voltage regulator for the downconversion of -12V to -5V. In the middle right we have the active band-pass filter and at the bottom we have the power-amp.

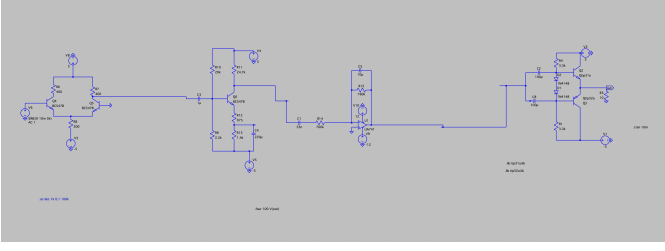


Fig. 1. LTSPICE Schematic of Full Circuit.

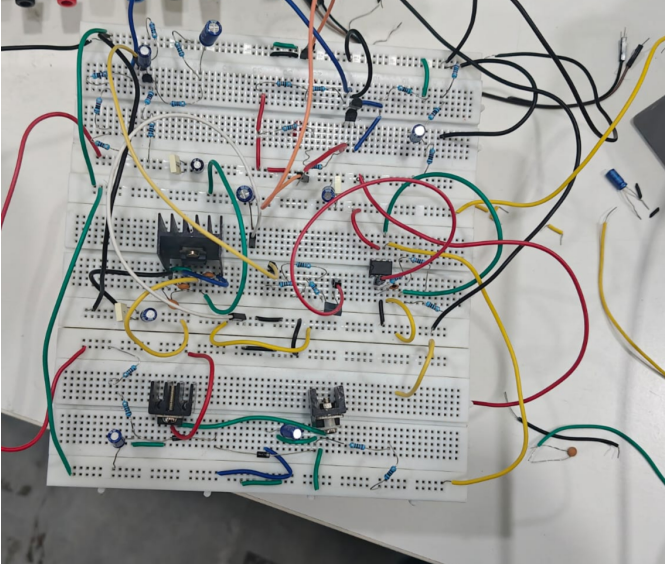


Fig. 2. Physically Implemented Circuit

IV. DIFFERENTIAL AMPLIFIER (PRE-AMPLIFIER STAGE)

A. Purpose

The differential amplifier provides initial amplification while attenuating common-mode noise via differential operation.

B. DC bias relationships (symbolic)

For a symmetric BJT differential pair with tail/emitter resistor R_E and (possibly) dual supplies, define the emitter node voltage V_E and base biases V_{B1}, V_{B2} . Under the usual approximation $|V_{BE}| \approx V_{BE,on}$,

$$I_E \approx \frac{V_E - V_{EE}}{R_E}, \quad I_C \approx \frac{I_E}{2} \quad (\text{matched pair}). \quad (2)$$

Then

$$g_m = \frac{I_C}{V_T}, \quad r_\pi \approx \frac{\beta}{g_m}. \quad (3)$$

C. Small-signal gain derivation

Let the collectors use equal resistors $R_{C1} = R_{C2} = R_C$. Using a small-signal model and defining the differential input

$$v_d = v_{in1} - v_{in2}, \quad (4)$$

the (single-ended) differential-mode output can be expressed as

$$v_{o1} = - \left(\frac{g_m R_C}{2} \right) v_d, \quad (5)$$

so the differential-mode gain is

$$A_d \triangleq \frac{v_{o1}}{v_d} = - \frac{g_m R_C}{2}. \quad (6)$$

If the input is driven symmetrically ($v_{in2} = -v_{in1}$), then $v_d = 2v_{in1}$ and the effective gain w.r.t. v_{in1} becomes

$$\frac{v_{o1}}{v_{in1}} = -g_m R_C. \quad (7)$$

D. Input/output impedance and CMRR

A first-order estimate of input resistance per input is

$$R_{in,1} \approx r_\pi \quad (\text{per transistor input}). \quad (8)$$

For single-ended output, the output resistance is approximately

$$R_{out,1} \approx R_C \parallel r_o \approx R_C \quad (r_o \gg R_C). \quad (9)$$

Define common-mode gain $A_c \triangleq v_{o1}/v_{cm}$ for $v_{in1} = v_{in2} = v_{cm}$. Then

$$\text{CMRR} = \frac{|A_d|}{|A_c|}, \quad \text{CMRR}_{dB} = 20 \log_{10} \left(\frac{|A_d|}{|A_c|} \right). \quad (10)$$

E. Pre-Gain Target Value Estimation

The differential pre-amplifier was designed with an initial target voltage gain of approximately 25 V/V (roughly 28 dB). This target was chosen to provide sufficient initial amplification to overcome the noise floor before the signal enters the high-gain common-emitter stage.

To reliably achieve this 25 V/V target in hardware, the circuit was deliberately over-designed. Knowing that real-world non-idealities—such as intrinsic transistor parasitic resistances (r_b , $R_{e,p}$) and the inevitable input impedance loading from the subsequent CE stage—would attenuate the signal, the component values (R_C and R_E) were explicitly chosen to yield a higher theoretical, unloaded gain of > 30 V/V.

By targeting a theoretical gain of approximately 33 V/V, the practically realized and loaded gain safely settled down to approximately 24 V/V to 27 V/V during testing. This intentional design margin successfully met the pre-gain estimation target while maintaining the transistors within their linear active region.

F. Practical implementation and gain calculation

In the practical implementation the differential amplifier was driven using a one sided input configuration. One transistor base receives the input signal while the other transistor base is grounded. This approach simplifies the experimental setup because it avoids the need to generate two input signals with an exact 180° phase difference. If the phase difference between the two inputs is not perfectly controlled, the signals can partially cancel each other which reduces the effective differential input.

Assuming the base of the grounded transistor is at 0 V and the base emitter voltage is approximately 0.7 V, the emitter node voltage is

$$V_E \approx -0.7 \text{ V}. \quad (11)$$

Since the emitter resistor is connected to $V_{EE} = -5 \text{ V}$, the tail current is

$$I_T = \frac{V_E - V_{EE}}{R_E} \quad (12)$$

1) *LTSPICE design values:* For the simulated circuit

$$R_E = 500 \Omega, \quad R_C = 400 \Omega \quad (13)$$

$$I_T = \frac{-0.7 - (-5)}{500} = \frac{4.3}{500} = 8.6 \text{ mA} \quad (14)$$

Assuming equal current sharing

$$I_C = \frac{I_T}{2} = 4.3 \text{ mA} \quad (15)$$

The transconductance is

$$g_m = \frac{I_C}{V_T} = \frac{4.3 \text{ mA}}{25.9 \text{ mV}} = 0.166 \text{ S} \quad (16)$$

For one sided excitation the gain becomes

$$A_{v1} \approx -\frac{g_m R_C}{2} \quad (17)$$

$$A_{v1} \approx -\frac{(0.166)(400)}{2} \quad (18)$$

$$A_{v1} \approx -33.2 \text{ V/V} \quad (19)$$

which corresponds to approximately

$$20 \log_{10}(33.2) \approx 30.4 \text{ dB}. \quad (20)$$

2) *Hardware values:* In the hardware implementation the closest standard resistor values were used

$$R_E = 510 \Omega, \quad R_C = 430 \Omega \quad (21)$$

$$I_T = \frac{4.3}{510} = 8.43 \text{ mA} \quad (22)$$

$$I_C = 4.22 \text{ mA} \quad (23)$$

$$g_m = \frac{4.22 \text{ mA}}{25.9 \text{ mV}} = 0.163 \text{ S} \quad (24)$$

$$A_{v1} \approx -\frac{(0.163)(430)}{2} \quad (25)$$

$$A_{v1} \approx -35.0 \text{ V/V} \quad (26)$$

which corresponds to approximately

$$20 \log_{10}(35.0) \approx 30.9 \text{ dB}. \quad (27)$$

3) Discrepancy Between Theoretical and Simulated Gain:

While the theoretical small-signal voltage gain was calculated as 33.2 V/V, the unloaded LTSPICE simulation yielded a gain of approximately 24 V/V. This discrepancy arises from the difference between the ideal first-order BJT model used in hand calculations and the more complex Gummel-Poon model employed by LTSPICE.

The theoretical calculation relies purely on the dynamic emitter resistance, given by $r_e = V_T/I_C$. However, the practical transistor model in LTSPICE includes physical parasitic resistances intrinsic to the device, namely the ohmic emitter resistance ($R_{e,p}$) and the base spreading resistance (r_b). These parasitics introduce unintentional emitter degeneration.

The effective small-signal emitter resistance in the simulation is:

$$r_{e,\text{eff}} = r_e + R_{e,p} + \frac{r_b}{\beta} \quad (28)$$

This increase in the effective emitter resistance proportionately reduces the effective transconductance ($g_{m,\text{eff}} \approx 1/r_{e,\text{eff}}$). Consequently, the voltage gain drops from the ideal textbook value down to the observed 24 V/V.

4) *Common-Mode Rejection Ratio (CMRR) Analysis:* For a differential amplifier with a single-ended output, the common-mode gain A_c is heavily dependent on the tail resistor R_E . It can be approximated as:

$$A_c \approx -\frac{R_C}{2R_E} \quad (29)$$

Using the simulation design values $R_C = 400 \Omega$ and $R_E = 500 \Omega$, the common-mode gain is:

$$A_c \approx -\frac{400}{2(500)} = -\frac{400}{1000} = -0.4 \text{ V/V} \quad (30)$$

With the previously calculated differential-mode theoretical gain of $|A_d| = 33.2 \text{ V/V}$, the linear Common-Mode Rejection Ratio is computed as:

$$\text{CMRR} = \frac{|A_d|}{|A_c|} = \frac{33.2}{0.4} = 83 \quad (31)$$

Expressing this value in decibels (dB) yields:

$$\text{CMRR}_{dB} = 20 \log_{10}(83) \approx 38.38 \text{ dB} \quad (32)$$

This confirms that the pre-amplifier stage effectively attenuates common-mode signals while providing the intended differential voltage gain.

G. LTSPICE schematic And Plots

Figure 3 shows the LTSPICE schematic of the differential amplifier used. Figure 4 shows the transient analysis output of the differential amplifier obtained from LTSPICE.

V. COMMON-EMITTER (CE) GAIN STAGE

A. Purpose

This stage provides the main voltage gain, while maintaining proper biasing in the active region.

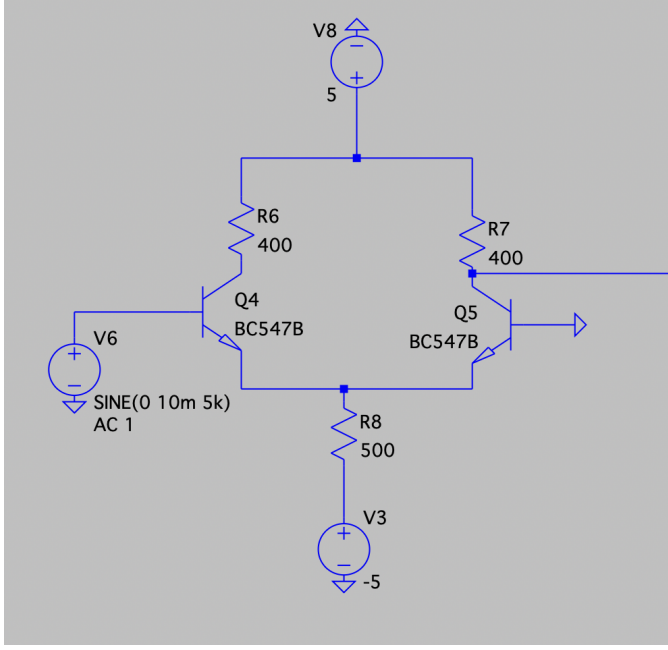


Fig. 3. LTSPICE schematic of the differential amplifier.

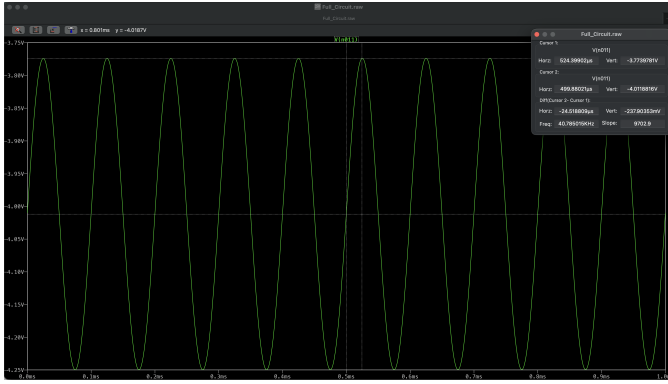


Fig. 4. LTSPICE transient response of the differential amplifier showing the differential output waveform.

B. DC Bias Design

Let the base be biased by a divider R_{B1}, R_{B2} from a supply V_{CC} (or from a Thevenin source). The base bias is

$$V_B = V_{CC} \frac{R_{B2}}{R_{B1} + R_{B2}} \quad (\text{single-supply divider form}). \quad (33)$$

With emitter resistors R_{E1} (bypassed for AC) and R_{E2} (unbypassed for AC), total DC emitter resistance is $R_E = R_{E1} + R_{E2}$ and

$$V_E \approx V_B - V_{BE,on}, \quad I_E \approx \frac{V_E - V_{EE}}{R_E}, \quad (34)$$

$$I_C \approx \frac{\beta}{\beta + 1} I_E \approx I_E, \quad g_m = \frac{I_C}{V_T}. \quad (35)$$

Collector DC level:

$$V_C = V_{CC} - I_C R_C. \quad (36)$$

Active-region check (symbolic) can be enforced by ensuring

$$V_{CE} = V_C - V_E > V_{CE,sat} \quad \text{and} \quad V_C > V_B \quad (\text{for NPN}). \quad (37)$$

C. Small Signal gain

Assuming coupling/bypass capacitors act as AC shorts in midband, and R_{E1} is bypassed while R_{E2} remains:

$$A_{v2} \triangleq \frac{v_o}{v_{in}} \approx -\frac{g_m(R_C \parallel R'_L)}{1 + g_m R_{E2}}, \quad (38)$$

where R'_L is the effective load seen at the collector (including the next stage input and any explicit load).

A commonly used equivalent form:

$$A_{v2} \approx -\frac{R_C \parallel R'_L}{\frac{1}{g_m} + R_{E2}}. \quad (39)$$

D. Active region verification

Using the calculated DC current $I_C \approx 0.121 \text{ mA}$ and $R_C = 24.7 \text{ k}\Omega$, the collector voltage is

$$V_C = V_{CC} - I_C R_C = 5 - (0.121 \text{ mA})(24.7 \text{ k}\Omega) \approx 2.01 \text{ V}. \quad (40)$$

Using $V_E \approx -4.71 \text{ V}$, the collector emitter voltage becomes

$$V_{CE} = V_C - V_E = 2.01 - (-4.71) \approx 6.72 \text{ V}. \quad (41)$$

Since V_{CE} is well above saturation, the transistor operates in the active region for the designed bias point.

E. LTSPICE validation of the CE gain stage

Figure 5 shows the LTSPICE schematic of the common-emitter (CE) gain stage used in the design. Figure 6 shows the LTSPICE transient response of the CE stage at 10 kHz, confirming stable amplification and correct waveform shape at the stage output.

F. Input impedance and impedance matching

The small signal input resistance is approximately

$$R_{in,2} \approx (R_{B1} \parallel R_{B2}) \parallel (r_\pi + (\beta + 1)R_{E2}). \quad (42)$$

For good inter-stage voltage transfer from the previous stage with output resistance $R_{out,1}$, require

$$R_{in,2} \gg R_{out,1}. \quad (43)$$

G. Practical biasing and gain calculation

The CE stage was biased using a resistor divider connected between the positive and negative supply rails.

$$R_{B1} = 20 \text{ k}\Omega, \quad R_{B2} = 2.2 \text{ k}\Omega \quad (44)$$

Using nodal analysis the base voltage is

$$V_B = \frac{R_{B2}(+5) + R_{B1}(-5)}{R_{B1} + R_{B2}} \quad (45)$$

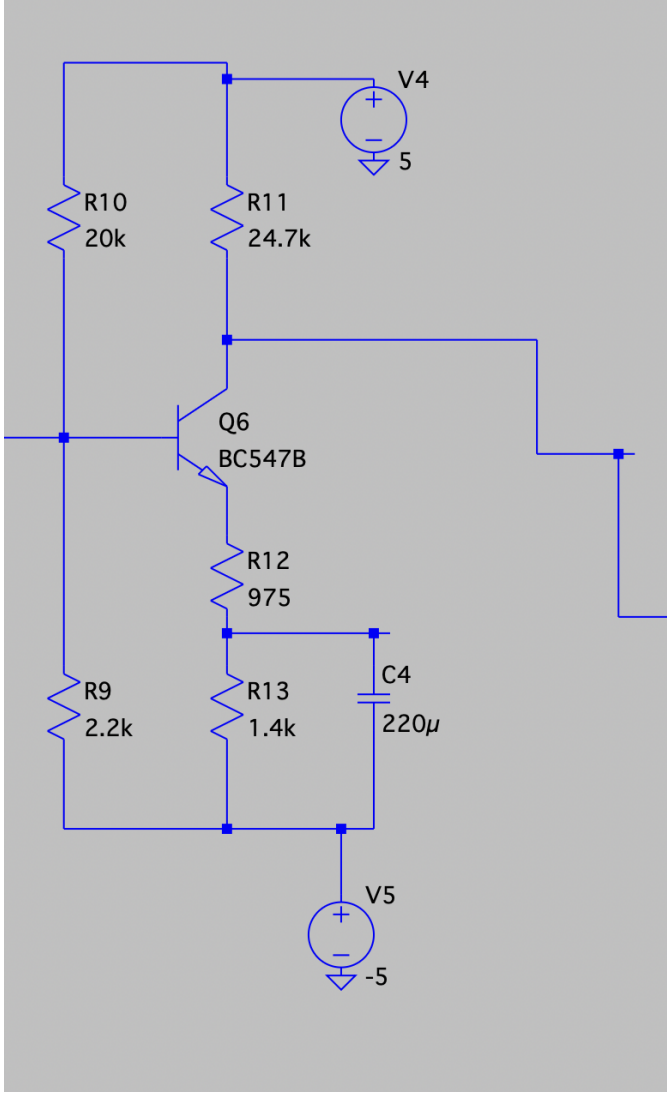


Fig. 5. LTSPICE schematic of the common-emitter amplifier (Stage 2).

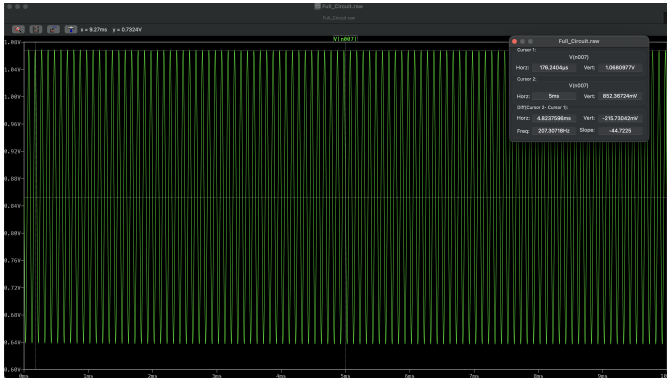


Fig. 6. LTSPICE transient output of the CE stage at 10 kHz.

$$V_B = \frac{(2.2k)(5) + (20k)(-5)}{22.2k} \quad (46)$$

$$V_B \approx -4.01 \text{ V} \quad (47)$$

The emitter voltage is therefore

$$V_E = V_B - V_{BE} \quad (48)$$

$$V_E = -4.01 - 0.7 = -4.71 \text{ V} \quad (49)$$

For DC analysis the bypass capacitor is open, so the emitter resistance is

$$R_{E,DC} = 1 \text{ k}\Omega + 1.4 \text{ k}\Omega = 2.4 \text{ k}\Omega \quad (50)$$

The emitter current becomes

$$I_E = \frac{V_E - V_{EE}}{R_E} \quad (51)$$

$$I_E = \frac{-4.71 - (-5)}{2.4k} \quad (52)$$

$$I_E = \frac{0.29}{2400} \quad (53)$$

$$I_E \approx 0.121 \text{ mA} \quad (54)$$

$$I_C \approx I_E \quad (55)$$

The transconductance becomes

$$g_m = \frac{I_C}{V_T} \quad (56)$$

$$g_m = \frac{0.121 \text{ mA}}{25.9 \text{ mV}} \quad (57)$$

$$g_m \approx 4.67 \times 10^{-3} \text{ S} \quad (58)$$

which gives

$$r_e = \frac{1}{g_m} \approx 214 \Omega \quad (59)$$

The collector resistor used was

$$R_C = 24.7 \text{ k}\Omega \quad (60)$$

1) *Reason for partial emitter bypass:* Two emitter resistors were used: 1 kΩ unbypassed and 1.4 kΩ bypassed using a 220 μF capacitor.

For DC operation both resistors determine the bias current and therefore stabilize the operating point.

For AC signals the capacitor provides a low impedance path at audio frequencies which effectively removes the 1.4 kΩ from the small signal emitter resistance. This increases the gain while still retaining some emitter degeneration for improved linearity.

2) *Voltage gain with bypass capacitor:* The effective emitter resistance for AC becomes

$$R_{E,AC} \approx 1 \text{ k}\Omega \quad (61)$$

The voltage gain is therefore approximately

$$A_{v2} \approx -\frac{R_C}{r_e + R_{E,AC}} \quad (62)$$

$$A_{v2} \approx -\frac{24700}{214 + 1000} \quad (63)$$

$$A_{v2} \approx -20.3 \text{ V/V} \quad (64)$$

which corresponds to

$$20 \log_{10}(20.3) \approx 26.2 \text{ dB}. \quad (65)$$

VI. ACTIVE BAND-PASS FILTER STAGE (f_L TO f_H)

A. Purpose

The filter should pass the audible band with minimal attenuation and suppress out-of-band components.

B. Transfer function derivation (symbolic)

Consider an inverting active band-pass structure with a high-pass input network (R_1, C_1) and low-pass feedback network (R_2, C_2). Define impedances

$$Z_1 = R_1 + \frac{1}{j\omega C_1}, \quad Z_2 = \frac{R_2}{1 + j\omega R_2 C_2}. \quad (66)$$

With an ideal op-amp, the inverting node is virtual ground ($v_- \approx 0$). By KCL at v_- :

$$\frac{0 - v_{in}}{Z_1} + \frac{0 - v_o}{Z_2} = 0 \Rightarrow \frac{v_o}{v_{in}} = -\frac{Z_2}{Z_1}. \quad (67)$$

Thus,

$$H(j\omega) \triangleq \frac{v_o}{v_{in}} = -\frac{R_2 j\omega C_1}{(1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2)}. \quad (68)$$

C. Cutoff frequencies and midband gain

The high-pass corner (lower cutoff) and low-pass corner (upper cutoff) are

$$f_L = \frac{1}{2\pi R_1 C_1}, \quad f_H = \frac{1}{2\pi R_2 C_2}. \quad (69)$$

In the midband region where $f_L \ll f \ll f_H$, the magnitude approaches

$$|H(j\omega)| \approx \frac{R_2}{R_1}. \quad (70)$$

For unity midband gain, set $R_2 = R_1$.

D. Cascading filter sections

If two identical stages are cascaded, the overall filter transfer is

$$H_{\text{total}}(j\omega) = (H(j\omega))^2, \quad (71)$$

and magnitude in dB adds (doubling the dB magnitude).

E. Chosen Component Values and Chosen Cutoffs

For the filter lower cutoff, we used $R = 760 \text{ k}\Omega$ and $C = 33 \text{ nF}$, giving

$$f_L \approx \frac{1}{2\pi(760 \times 10^3)(33 \times 10^{-9})} \approx 6.34 \text{ Hz}. \quad (72)$$

For the filter upper cutoff, we used $R = 760 \text{ k}\Omega$ and $C = 10 \text{ pF}$, giving

$$f_H \approx \frac{1}{2\pi(760 \times 10^3)(10 \times 10^{-12})} \approx 2.094 \times 10^4 \text{ Hz} (\approx 20.94 \text{ kHz}). \quad (73)$$

The filter was designed with a conservative lower cutoff of approximately 6 Hz so that the entire 20 Hz audio band can pass without attenuation. However, at the system level the effective lower cutoff frequency shifts upward to approximately 145 Hz due to the presence of multiple interstage coupling and decoupling capacitors, which introduce additional high-pass filtering effects in the overall amplifier chain.

F. LTSPICE Validation of Filter Stage

Figure 8 shows the LTSPICE AC response of the filter stage. In the operational region, the gain is approximately 0 dB as expected for a unity gain filter. Figure 7 showing the the LTSPICE schematic of the Band Pass Filter .

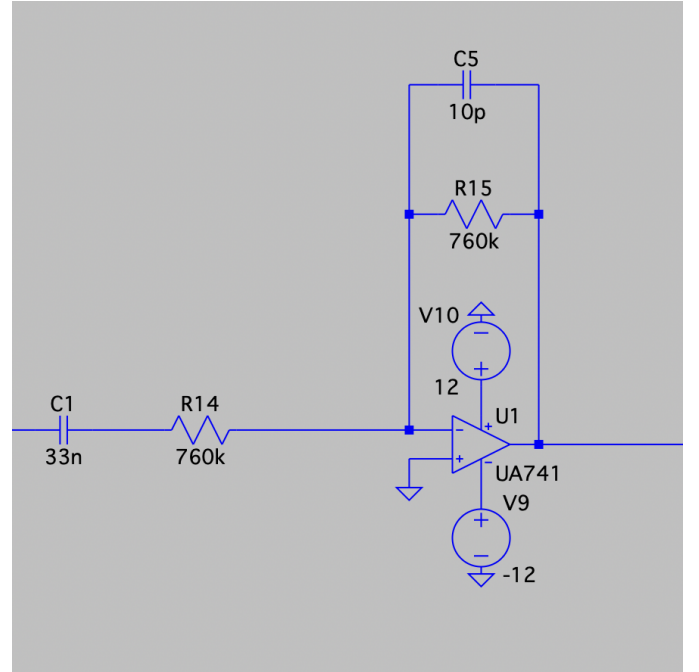


Fig. 7. LTSPICE schematic of the Band Pass Filter.

VII. POWER AMPLIFIER STAGE (CLASS-AB)

A. Purpose

The final stage provides current drive into a low-resistance load while maintaining approximately unity voltage gain.

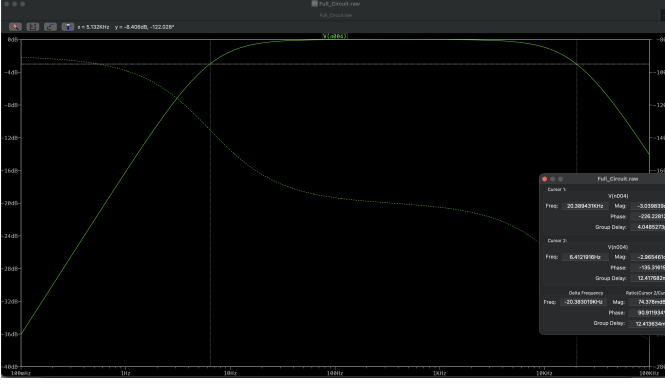


Fig. 8. LTSPICE Bode plot of the filter stage confirming unity gain in the passband.

B. Class A, Class B, and Class AB

In a class-A amplifier, the active device conducts for the entire cycle (360°). This is linear but inefficient because the device dissipates significant power even with no signal.

In a class-B push-pull stage, each device conducts for approximately half the cycle (180°). This improves efficiency, but near the zero crossing there is a dead-zone due to the turn-on voltage (e.g., V_{BE}), causing crossover distortion.

Class-AB biases both devices slightly ON, so conduction is more than 180° per device. This greatly reduces crossover distortion while keeping efficiency much better than class-A. Since our amplifier is intended for audio, we chose class-AB to get cleaner output than class-B without the thermal penalty of class-A.

C. Biasing condition (symbolic)

For complementary emitter followers, assume $|V_{BE}| \approx V_{BE,on}$. The required bias between the two bases is

$$V_{B+} - V_{B-} \approx 2V_{BE,on}. \quad (74)$$

A diode string or VBE-multiplier can set this separation to reduce crossover distortion.

D. Small-signal gain (unity constraint)

Approximating matched devices ($g_{mN} \approx g_{mP} \approx g_m$, $r_{\pi N} \approx r_{\pi P} \approx r_\pi$), a compact form for the emitter-follower attenuation into R_L is

$$A_{v4} \triangleq \frac{v_o}{v_{in}} \approx \frac{x}{x + \frac{1}{R_L}}, \quad x \triangleq \left(\frac{1}{r_{\pi N}} + \frac{1}{r_{\pi P}} + g_{mN} + g_{mP} \right) \quad (75)$$

To ensure $A_{v4} \approx 1$ (no additional voltage gain and minimal attenuation), require

$$x \gg \frac{1}{R_L}. \quad (76)$$

E. Power Amplifier Bias Current and Output Power Calculations

Biasing was implemented using two 1N4148 diodes between the bases, with two resistors of $R = 3.3 \text{ k}\Omega$ connected

from the supply rails to the ends of the diode pair. The bias current through the diode string is approximated as

$$I_{bias} \approx \frac{V_{CC} - V_{EE} - 2V_D}{2R}. \quad (77)$$

Using $V_{CC} = 5 \text{ V}$, $V_{EE} = -5 \text{ V}$, and $V_D \approx 0.7 \text{ V}$,

$$I_{bias} \approx \frac{10 - 1.4}{6.6 \text{ k}} \approx 1.30 \text{ mA}. \quad (78)$$

For a 10Ω load, if the output is $v_o(t) = V_m \sin(2\pi ft)$, then

$$V_{rms} = \frac{V_m}{\sqrt{2}} \quad (79)$$

and the power delivered to the load is

$$P_{out} = \frac{V_{rms}^2}{R_L} = \frac{V_m^2}{2R_L}. \quad (80)$$

With $R_L = 10 \Omega$,

$$P_{out} = \frac{V_m^2}{20}. \quad (81)$$

F. Load current requirement and device selection

For a load R_L , the RMS output voltage required to deliver power P_{out} is

$$V_{rms} = \sqrt{P_{out} R_L}. \quad (82)$$

For $P_{out} = 1.5 \text{ W}$ and $R_L = 10 \Omega$,

$$V_{rms} = \sqrt{15} \approx 3.87 \text{ V}. \quad (83)$$

G. LTSPICE Schematic And Unity gain verification

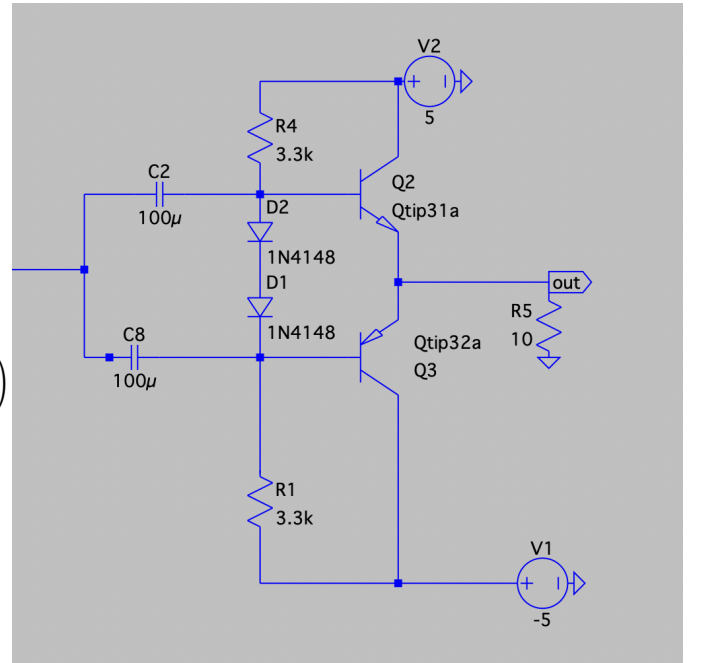


Fig. 9. LTSpice Schematic of Power Amp.

The power stage primarily provides current gain. The voltage gain

$$A_{v4} = \frac{V_{out}}{V_{in,4}} \quad (84)$$

was verified at 10 kHz and found to be approximately unity, as expected for an emitter follower configuration.

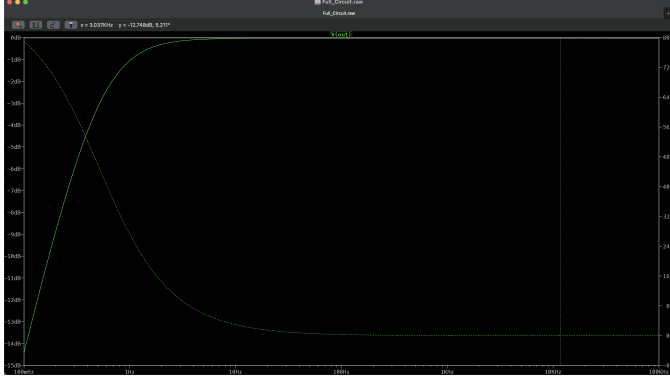


Fig. 10. LTspice AC magnitude of the power stage showing approximately unity gain.

VIII. OVERALL PERFORMANCE & ANALYSIS

A. Total gain and verification

The overall midband voltage gain of the amplifier is the product of the gains of the individual stages

$$A_v^{\text{total}} = A_{v1} A_{v2} A_{v3} A_{v4}. \quad (85)$$

From simulation and hardware measurements, the approximate gains of the stages are

- Differential pre-amplifier: $A_{v1} \approx 24$
- CE gain stage: $A_{v2} \approx 21$
- Active filter stage: $A_{v3} \approx 1$
- Power amplifier stage: $A_{v4} \approx 1$

Substituting these values,

$$A_v^{\text{total}} \approx (24)(21)(1)(1) \quad (86)$$

$$A_v^{\text{total}} \approx 504 \text{ V/V}. \quad (87)$$

Expressed in decibels,

$$A_v^{\text{total}}(\text{dB}) = 20 \log_{10}(504) \approx 54.05 \text{ dB}. \quad (88)$$

This value closely matches the measured steady-state gain from the Bode plot of the complete amplifier, which was observed to be approximately 52.3 dB. The small difference arises due to loading effects between stages, transistor parameter variations, and non ideal behaviour of practical components.

B. LTspice validation of complete amplifier

To verify the complete signal chain in simulation, both transient and AC analyses were performed in LTspice on the full amplifier.

1) *LTspice transient response of complete amplifier:* A transient analysis was run with a sinusoidal input and the output was observed at the final node after the power amplifier stage. Figure 11 shows the input and final output waveforms of the complete amplifier. The output remains sinusoidal at the test frequency and the amplitude increase is consistent with the expected overall gain.

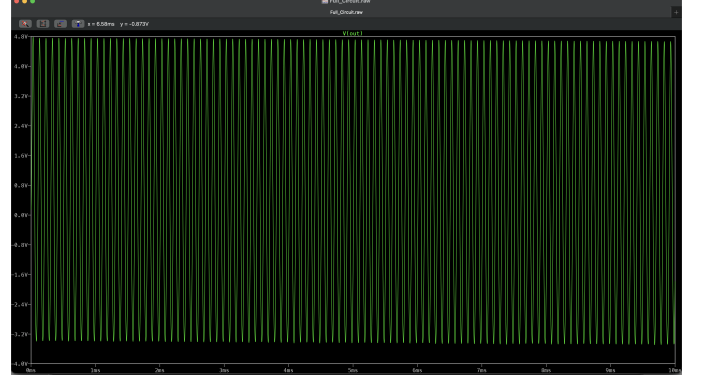


Fig. 11. LTspice transient response of the complete amplifier showing input and final output.

2) *LTspice AC response of complete amplifier:* An AC sweep was performed on the complete amplifier to verify the overall frequency response and bandwidth. Figure 12 shows the magnitude response of the complete amplifier from LTspice.

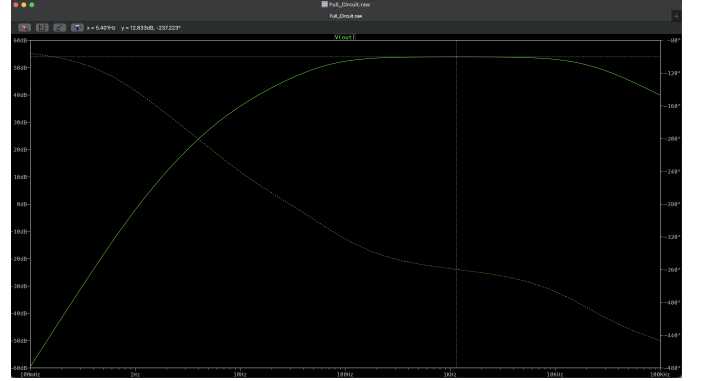


Fig. 12. LTspice AC magnitude response of the complete amplifier.

C. Total Harmonic Distortion

Total Harmonic Distortion quantifies nonlinear distortion by comparing the RMS contribution of the harmonics to the RMS value of the fundamental. In general,

$$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} V_{n,\text{rms}}^2}}{V_{1,\text{rms}}} \times 100\%. \quad (89)$$

1) *LTSPICE THD using Fourier analysis*: LTSPICE provides THD directly using Fourier analysis on the output node (for example using the `.four` directive). For the operating point used in the LTSPICE run shown in the Fourier log, the fundamental was at 1 kHz and the simulator reported

$$\text{THD}_{\text{LTspice}} \approx 7.81\%. \quad (90)$$

In our LTspice runs, this higher THD is mainly influenced by the exact transistor models, operating point, and the fact that simulation assumes ideal wiring and component interconnects while still including device level nonlinearities. Practical hardware measurements can show lower THD when the bias point, loading, and actual signal swing are slightly different from the simulation setup.

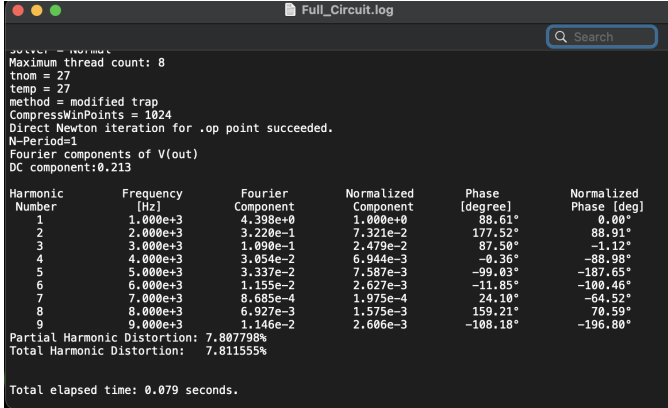


Fig. 13. LTSPICE Fourier log for the complete amplifier output $V(out)$.

2) *Hardware THD using FFT readings*: From the oscilloscope FFT, the harmonic levels were recorded in dBV. If the n th harmonic and the fundamental are given in dBV as dBV_n and dBV_1 , the amplitude ratio is

$$\frac{V_n}{V_1} = 10^{\frac{\text{dBV}_n - \text{dBV}_1}{20}}. \quad (91)$$

The THD estimate using harmonics up to N is

$$\text{THD} \approx 100 \sqrt{\sum_{n=2}^N \left(\frac{V_n}{V_1} \right)^2}. \quad (92)$$

Using the measured FFT values

- 1 kHz: 6.25 dBV
- 2 kHz: -25 dBV
- 3 kHz: -33.125 dBV
- 4 kHz: -48 dBV
- 5 kHz: -53 dBV

hence on using the formula, the corresponding THD up to the 5th harmonic is approximately

$$\text{THD}_{\text{HW}} \approx 2.95\%. \quad (93)$$

In our hardware measurements, the observed THD was around 2%. The exact value depends on the chosen measurement bandwidth, the number of harmonics included, the input

amplitude, and the oscilloscope FFT settings. Both values are consistent with low distortion operation for this lab scale audio amplifier.

D. Slew rate requirement and verification

For a sinusoidal output $v_o(t) = V_m \sin(2\pi ft)$, the maximum slope occurs at the zero crossing:

$$\left(\frac{dv_o}{dt} \right)_{\max} = 2\pi f V_m. \quad (94)$$

Therefore, the minimum required slew rate to reproduce a sine wave of peak amplitude V_m at frequency f without slew induced distortion is

$$\text{SR}_{\text{req}} = 2\pi f V_m. \quad (95)$$

1) *LTspice measured slew rate*: To estimate the slew rate in LTspice, a transient simulation was performed and the output slope was extracted by plotting the derivative $d(V(out))$. The peak magnitude of this derivative corresponds directly to the slew rate of the amplifier.

Figure 14 shows the plotted derivative $d(V(out))$ used for extracting the maximum slope.

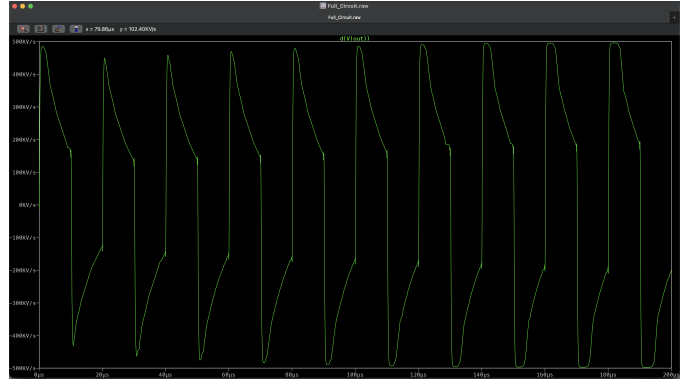


Fig. 14. LTspice plot of $d(V(out))$ used to estimate the slew rate. The maximum magnitude of this curve gives the slew rate.

From the $d(V(out))$ waveform, the maximum measured slope at the output was approximately

$$\text{SR}_{\text{LTspice}} \approx 450 \text{ kV/s}. \quad (96)$$

Converting units,

$$450 \text{ kV/s} = 450000 \text{ V/s} = 0.45 \text{ V}/\mu\text{s}. \quad (97)$$

This slope is not too large by itself. It only becomes a problem if the required slope at the output for the intended maximum frequency and amplitude exceeds what the circuit can deliver. In our operating range, the measured slope indicates that the circuit is not slew limited for moderate output amplitudes.

2) *Amplitude limit near the upper cutoff:* At the measured upper cutoff frequency $f_{\max} \approx 22.15$ kHz, the maximum peak amplitude that can be supported without slew limiting is

$$V_{m,\max} \approx \frac{SR_{LTspice}}{2\pi f_{\max}} = \frac{450000}{2\pi(22150)} \approx 3.23 \text{ V}. \quad (98)$$

If the output peak amplitude is significantly above this value near 22 kHz, slew rate limiting can contribute to distortion and increased harmonic content.

IX. HARDWARE MEASUREMENTS (OSCILLOSCOPE RESULTS)

A. Transient Response at 10 kHz

To verify signal propagation through the amplifier chain, transient measurements were taken at an input frequency of 10 kHz. The outputs of each stage were observed sequentially to confirm correct amplification and signal integrity.

The output of the differential amplifier (Stage 1) is shown in Fig. 15. This stage performs the initial amplification while suppressing common-mode noise.

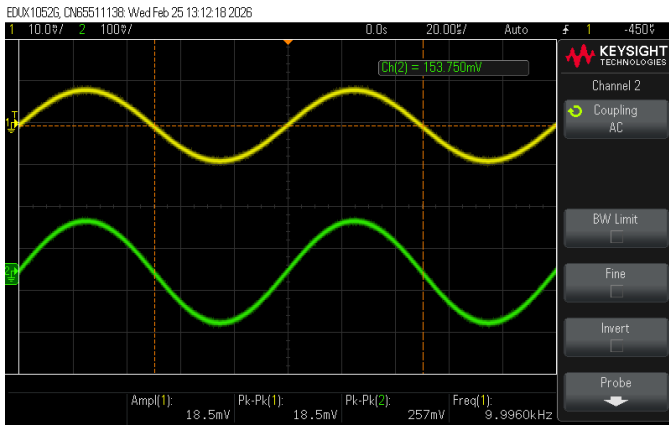


Fig. 15. Output waveform of the differential amplifier (Stage 1) at 10 kHz.

The output of the common-emitter voltage gain stage (Stage 2) is shown in Fig. 16. This stage provides the majority of the voltage gain in the amplifier.

The signal after the active band-pass filter (Stage 3) is shown in Fig. 17. The filter stage limits the bandwidth of the amplifier and ensures that only signals within the intended frequency range are passed.

The final output of the amplifier after the power stage (Stage 4) is shown in Fig. 18. The output stage provides current amplification and drives the load while maintaining approximately unity voltage gain.

B. Frequency Response

The overall frequency response of the complete amplifier is shown in Fig. 19. Within the operational region of the amplifier, the steady-state gain is observed to be approximately 52.3 dB. The measured lower cutoff frequency occurs around 145 Hz, which is significantly higher than the theoretical

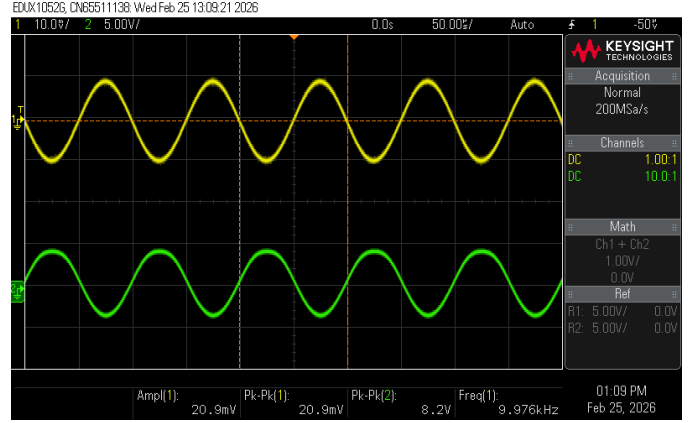


Fig. 16. Output waveform of the CE voltage gain stage (Stage 2) at 10 kHz.

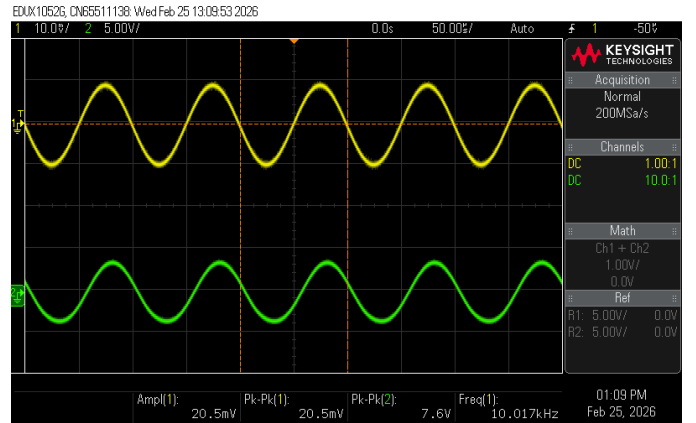


Fig. 17. Output waveform of the active band-pass filter (Stage 3) at 10 kHz.

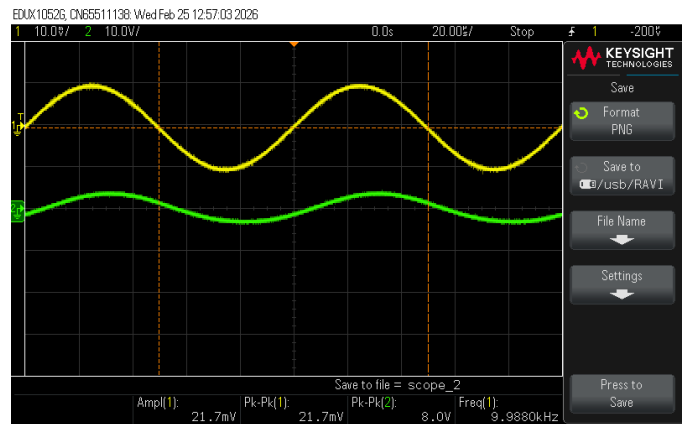


Fig. 18. Final output waveform after the class-AB power amplifier (Stage 4) at 10 kHz.

design value of approximately 6 Hz. This shift is primarily due to the presence of multiple coupling and decoupling capacitors inserted throughout the amplifier stages. These capacitors were required to ensure proper bias isolation and stable operation between stages, but they introduce additional high-pass filtering effects that raise the effective lower cutoff frequency.

The measured upper cutoff frequency occurs at approximately 22.15 kHz, which closely matches the expected design value derived from the filter component selection. This confirms that the upper frequency limit of the amplifier is mainly determined by the designed filter stage.

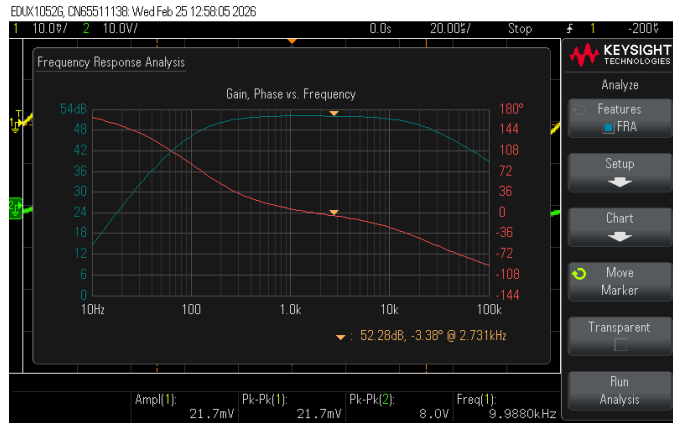


Fig. 19. Measured Bode plot of the complete amplifier.

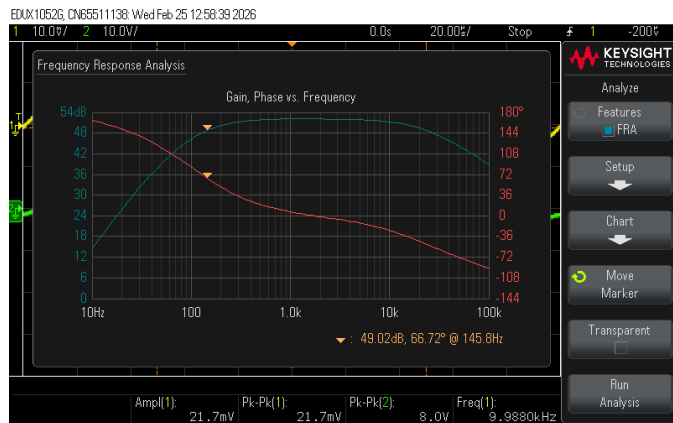


Fig. 20. Measured lower cutoff frequency of the complete amplifier.

C. Filter Stage Frequency Response

The frequency response of the filter stage alone is shown in Fig. 22 and Fig. 23. In the passband region, the steady-state gain is observed to be approximately 0 dB, which is consistent with the design of a unity-gain active band-pass filter.

Similar to the full amplifier response, the lower cutoff frequency appears near 145 Hz. Again, this deviation from the theoretical filter design (approximately 6 Hz) arises from the coupling and decoupling capacitors used between stages, which collectively introduce additional high-pass behavior.

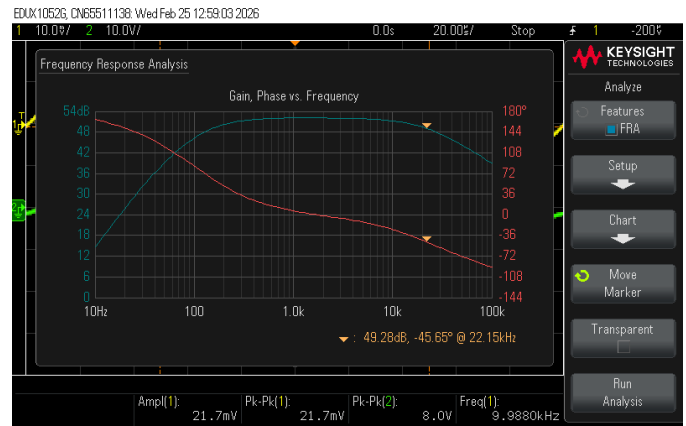


Fig. 21. Measured upper cutoff frequency of the complete amplifier.

The upper cutoff frequency of the filter stage is measured at approximately 22.15 kHz. This value matches the upper cutoff observed in the complete amplifier response, indicating that the filter stage primarily determines the high-frequency roll-off of the overall system.

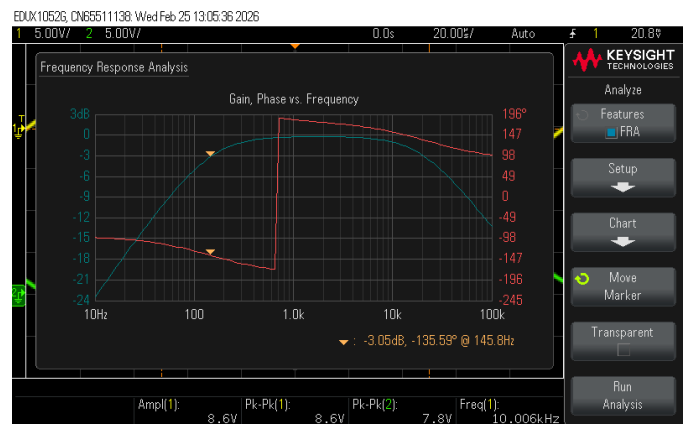


Fig. 22. Measured lower cutoff frequency of the filter stage.

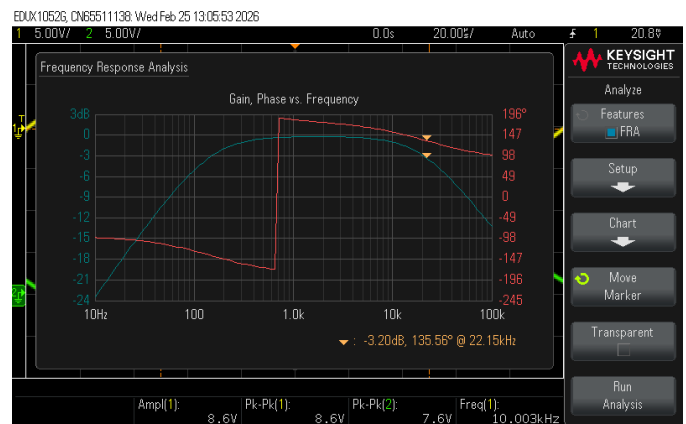


Fig. 23. Measured upper cutoff frequency of the filter stage.

D. FFT Analysis

The harmonic content of the output signal was analyzed using the FFT feature of the oscilloscope. The resulting spectrum is shown in Fig. 24.

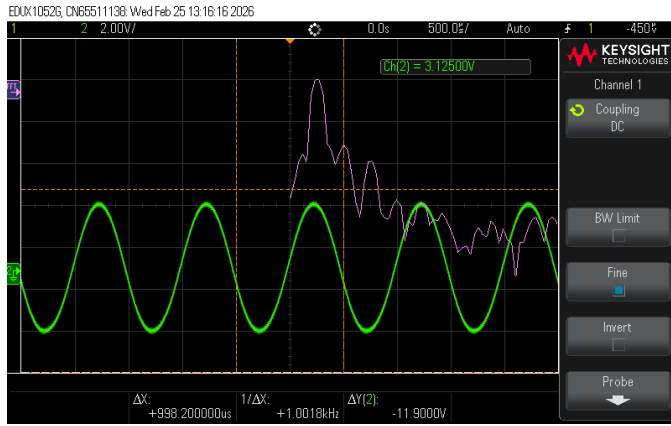


Fig. 24. FFT spectrum of the amplifier output used for harmonic distortion analysis.

X. CONCLUSION

A four-stage analog audio amplifier was successfully designed, simulated, and implemented in hardware. The system consists of a differential pre-amplifier for initial gain and common-mode noise rejection, a common-emitter voltage gain stage providing the majority of the voltage amplification, an active band-pass filter shaping the audio bandwidth, and a class-AB output stage for current amplification and load driving capability.

The measured overall midband gain of the amplifier was approximately 52.3 dB, which is consistent with the expected gain derived from the individual stage gains. The measured frequency response shows a lower cutoff near 145 Hz and an upper cutoff near 22.15 kHz. While the theoretical lower cutoff of the filter stage was designed around 6 Hz, the higher observed cutoff frequency results from additional high-pass effects introduced by interstage coupling capacitors.

Total harmonic distortion was evaluated both in simulation and in hardware. LTspice Fourier analysis reported approximately 7.81% THD, while oscilloscope FFT measurements indicated a lower distortion level of approximately 2% in the practical implementation.

The slew rate of the amplifier was estimated from LTspice transient analysis as approximately $0.45 \text{ V}/\mu\text{s}$, which is sufficient to reproduce audio signals within the operating bandwidth without significant slew-rate induced distortion for moderate output amplitudes.

Overall, the designed amplifier meets the intended functional requirements and demonstrates the successful integration of differential amplification, high-gain transistor stages, active filtering, and class-AB power amplification in a complete analog signal chain.

REFERENCES

- [1] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed., Oxford Univ. Press, 2014.
- [2] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed., Cambridge Univ. Press, 2015.